Bilkent University EE313 ECD Lab 4 Final Work Report

Wide-Band Amplifier with Feedback

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Introduction and Aim

In this experiment, our aim was to design a two stage wide-band amplifier with feedback to achieve low output impedance and a flat gain. I used 2 BJTs in my design. Here are some given specifications of the experiment:

- Source impedance: 500Ω
- Load impedance: 47Ω
- Mid band voltage Gain: 20 dB±0.5dB
- Bandwidth (−3dB): at least 2KHz to 2MHz (by CNTL-Click in AC analysis)
- Supply voltage: 12V (single supply)
- Maximum current consumption: 70mA from the supply voltage
- Undistorted peak-to-peak output voltage: 2Vpp at 200KHz.
- Distortion at the output: Harmonics less than −30dBc at 200KHz 2Vpp output voltage (the difference between the fundamental and the highest harmonic in FFT window)

Software Part

Specifications and Design

I used BC238 for my BJTs. I designed according to the given circuit diagram to us. Here is my full design of the circuit and the circuit diagram:

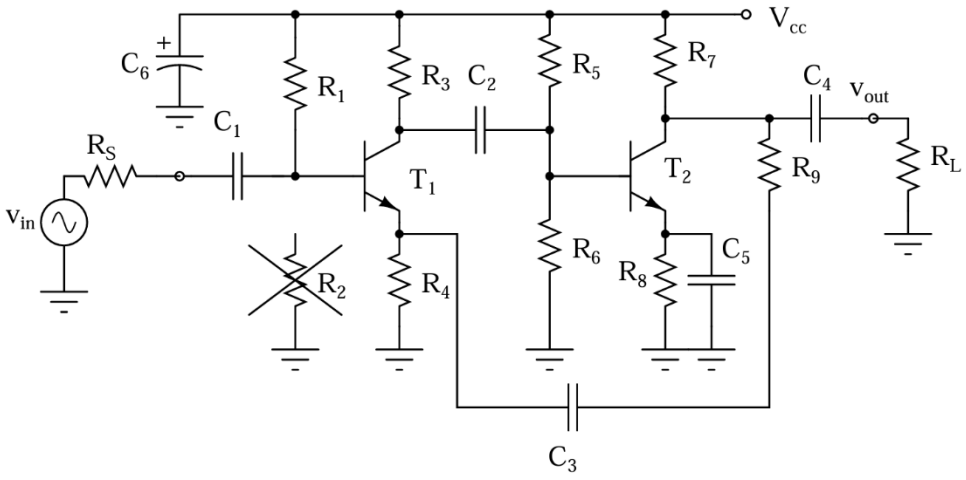


Figure 1: Followed circuit design.

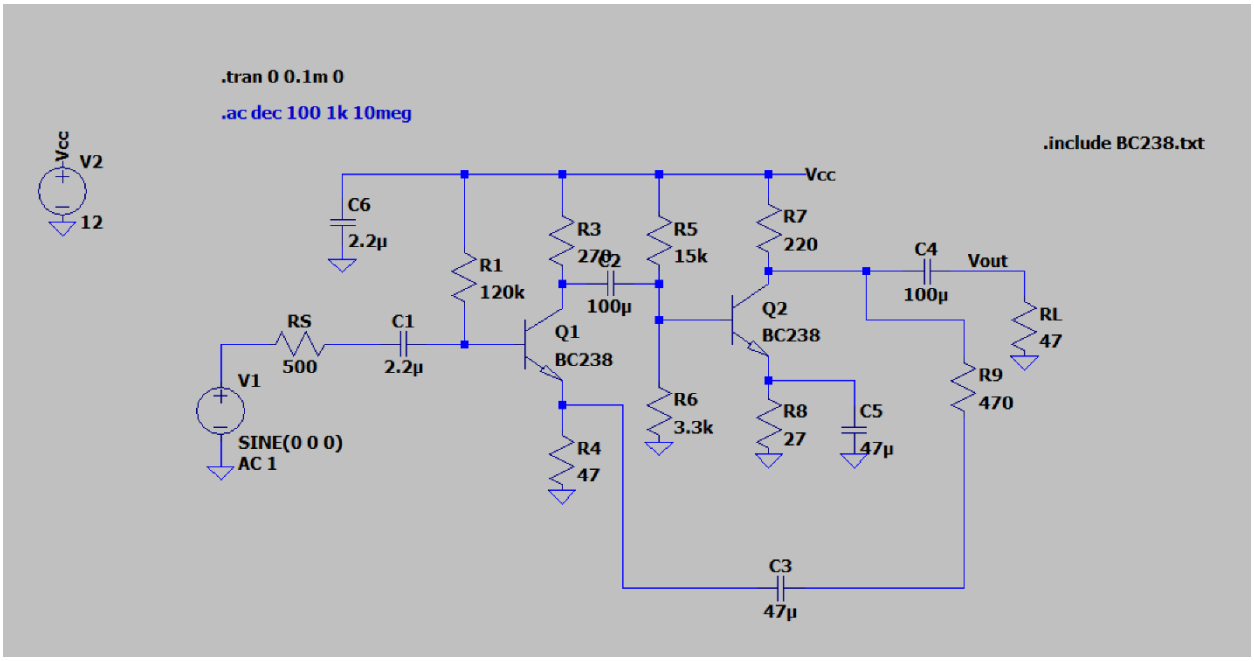


Figure 2: Designed circuit on LTSpice.

The values of the components are chosen according to the lab manual. I chose R4 as 47Ω and R8 as 27Ω to make the voltages around them 1V. To bias the BJTs, R1, R5, and R6 are chosen according to the manual. These values are chosen carefully to make the currents passing on these transistors correctly with the standard resistor values. To make sure that BJTs are always in active region, R3 and R7 are chosen accordingly and also, to make the gain as large as possible. For R7:

$$V_{CC} - V_{c2} = 7.7V, and\ the\ current\ on\ that\ resistor\ is\ 0.037A.$$

$$Therefore\ we\ can\ calculate\ R7$$

We can calculate R3 likewise. They are chosen as 270Ω and 220Ω respectively. After that, since the voltages on R4 and R8 must be 1V for BETA stability and the currents on Q1 and Q2 must be lower than 70mA in total we can calculate these resistor values. The current on Q1 is chosen around 24mA and the current on Q2 is around 37mA. To calculate R4 and R8 we can use the formulas below.

$$0.024 * R_4 \approx 1V \; and \; 0.037 * R_8 \approx 1V$$

To make this equation hold for standard values of resistors, R_4 is chosen as 47Ω and R_8 is chosen as 27Ω. I chose $R_3 = 270\Omega$ to make sure that our BJT is always in ACT region.

Since R4 is chosen as 47Ω, the value R9 is chosen as 10 times of that value because of the gain we should get which is 20dB (10 times). The capacitor values are chosen according to the instructions given to us as enough DC block capacitors.

Now for the specifications:

1. The current consumption is less than 70mA.

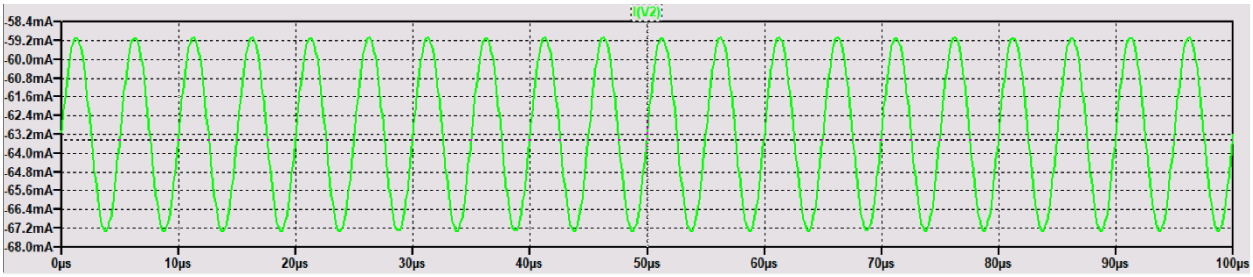


Figure 3: Current Consumption Plot

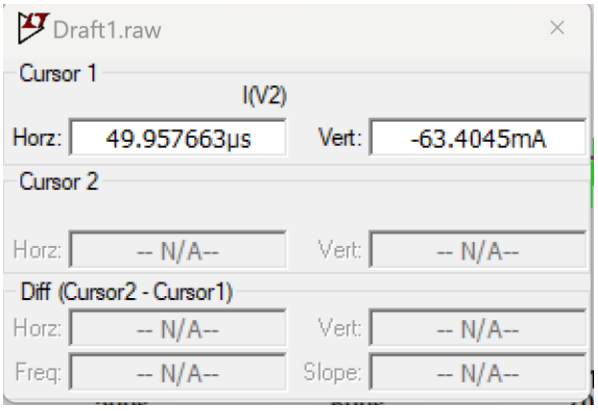


Figure 4: Current Consumption Plot Cursor

As it can be seen on the plot, the current consumption is less than 70mA, therefore this requirement is satisfied.

2. The small-signal bandwidth is at least 2KHz-2MHz while the mid-band gain is 20dB±0.5dB (by AC analysis).

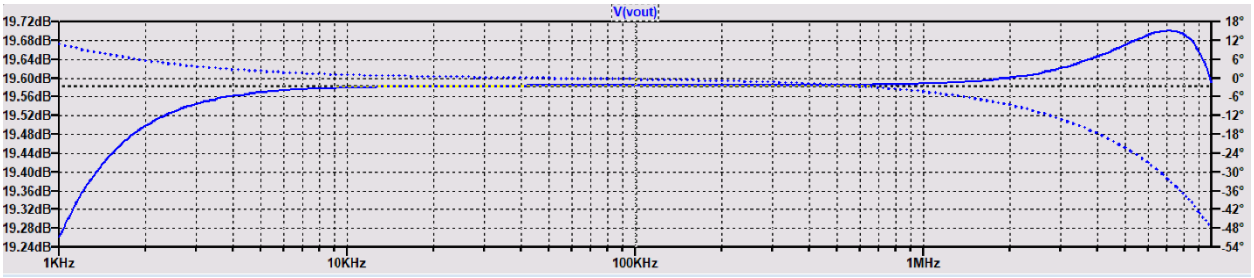


Figure 5: Gain vs freq plot.

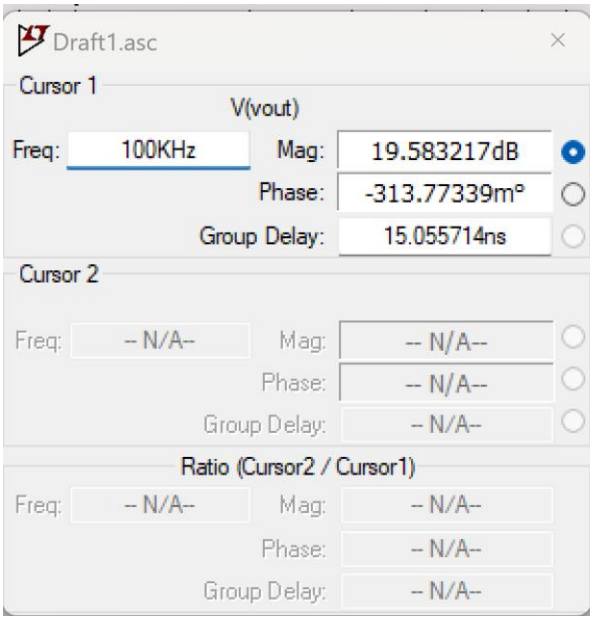


Figure 6: Cursor

As it can be seen from the plot, the mid-band gain is around 19.58dB which is in our boundaries 20dB±0.5dB. Therefore, this requirement is also satisfied.

3. The harmonic content of the output voltage is better than −30dBc with 0.1V peak input signal at 200KHz.

Firstly, here is the output signal which is around 2Vpp, an undistorted sinusoidal signal. I chose to look at the difference between the first 2 harmonics at 200kHz and 400kHz.

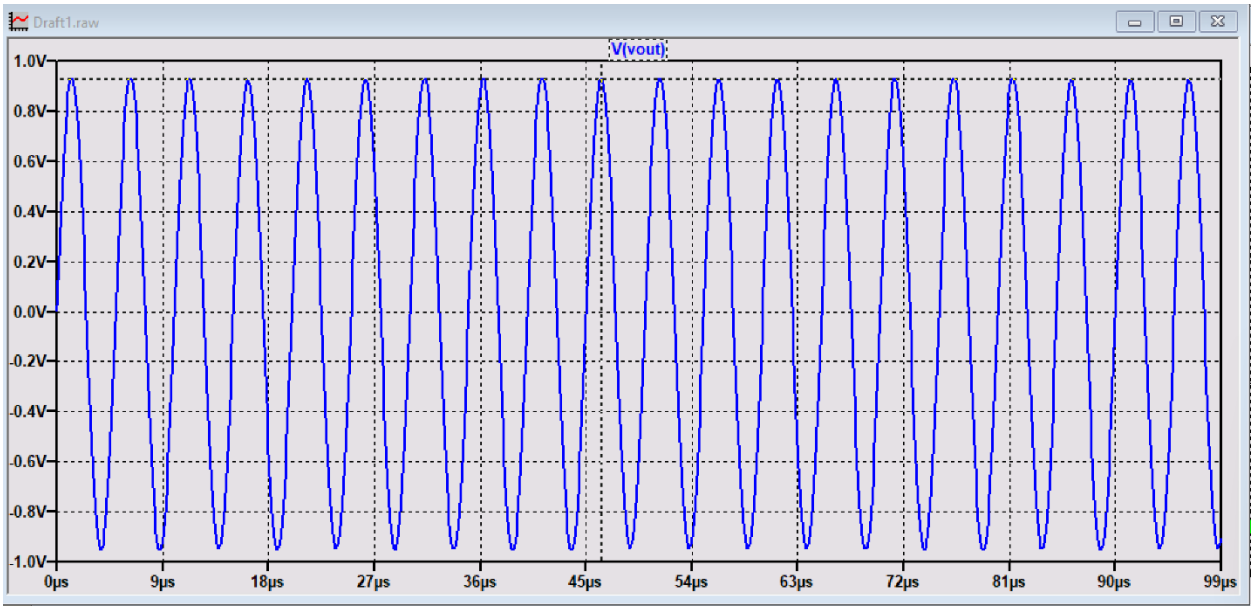


Figure 7: Vout

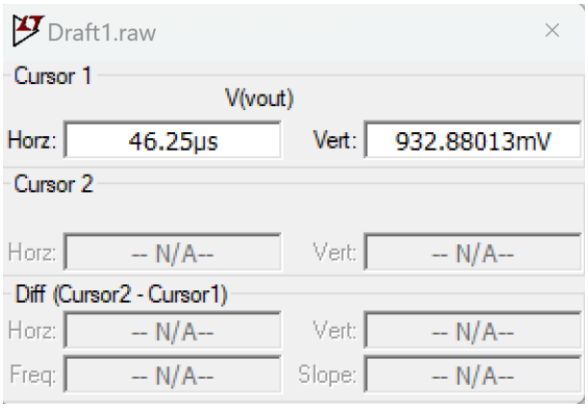


Figure 8: Vout

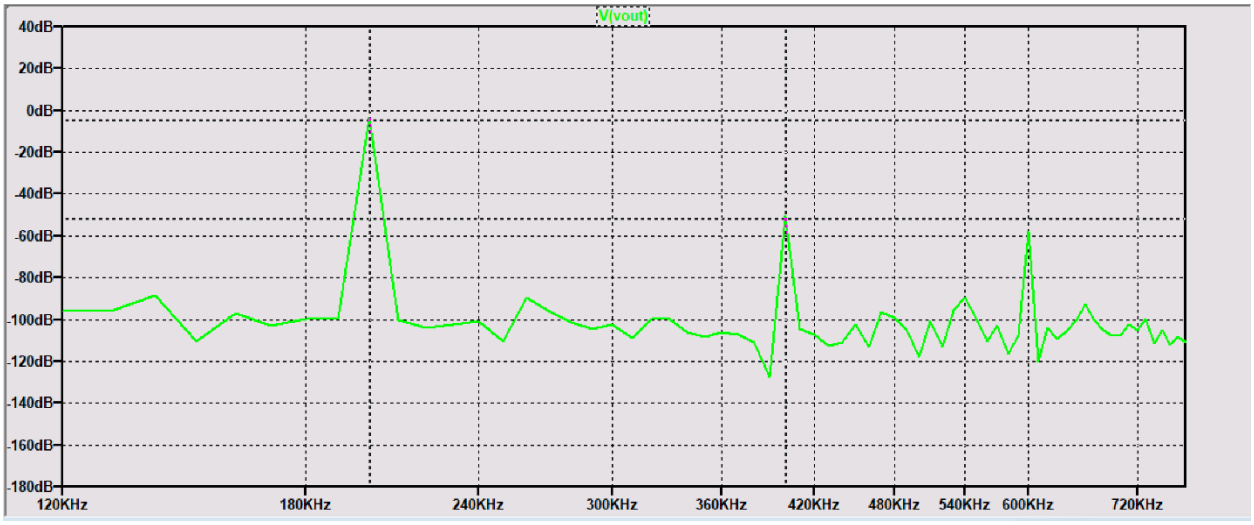


Figure 9: Plot of harmonics.

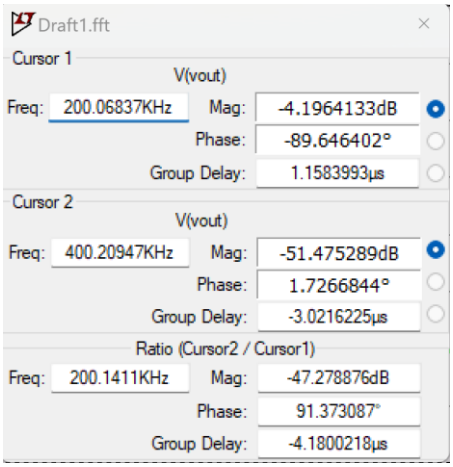


Figure 10: Difference between the harmonics.

As it can be seen on the graphs, the harmonic content difference is around 47.27dB which is better than 30dB. Therefore, this requirement is also satisfied.

4. The small-signal input impedance of the amplifier at 200KHz (the adjusted value of R_S. in AC analysis until the voltage gain drops by 6dB compared to R_S. =0).

To check this, I firstly shorted R_S. Then I checked the gain.

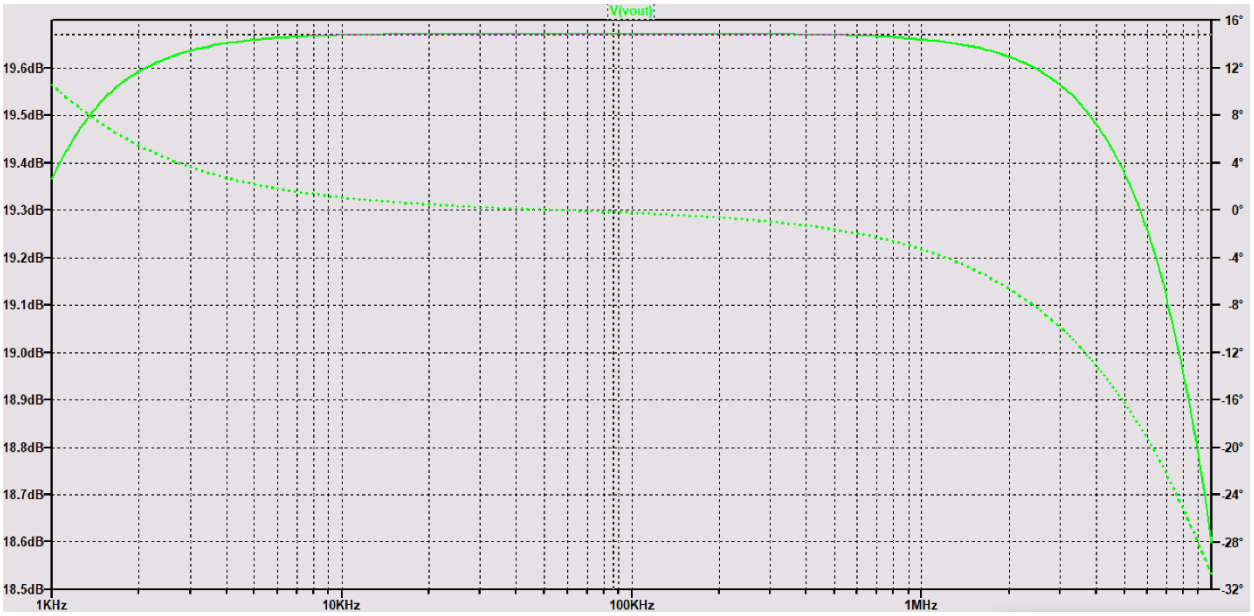


Figure 11: Gain plot when RS is shorted.

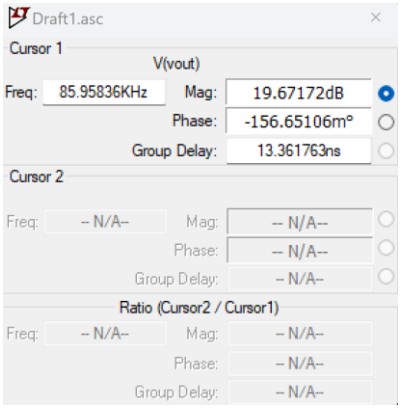


Figure 12: Gain value when RS is shorted.

Then when $R_S = 47\Omega$,

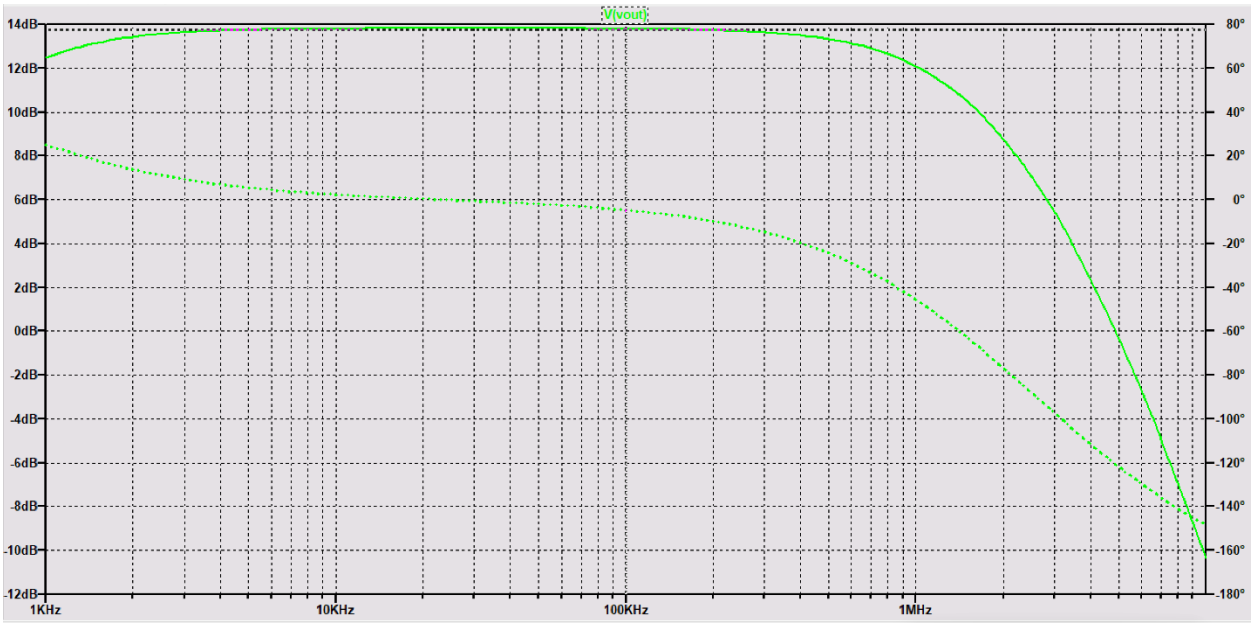


Figure 13: Gain plot when Rs=47Ω

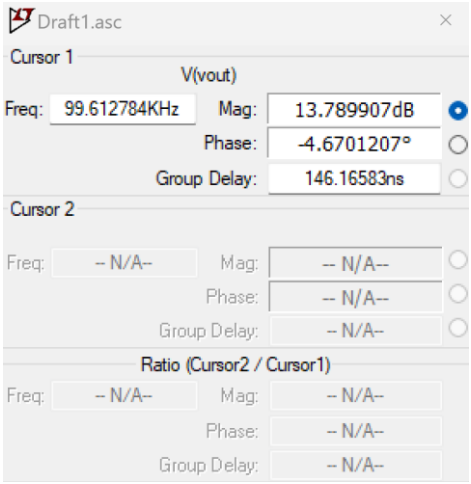


Figure 14: Gain value when Rs=47Ω.

As it can be seen from the plots, we almost reached a drop of 6dB when the resistor value is 47Ω.

5. The small-signal output impedance of the amplifier at 200KHz (the adjusted value of RL in AC analysis until the voltage gain drops by 6dB compared to RL=∞)

To make this, I open-circuited the load resistor.

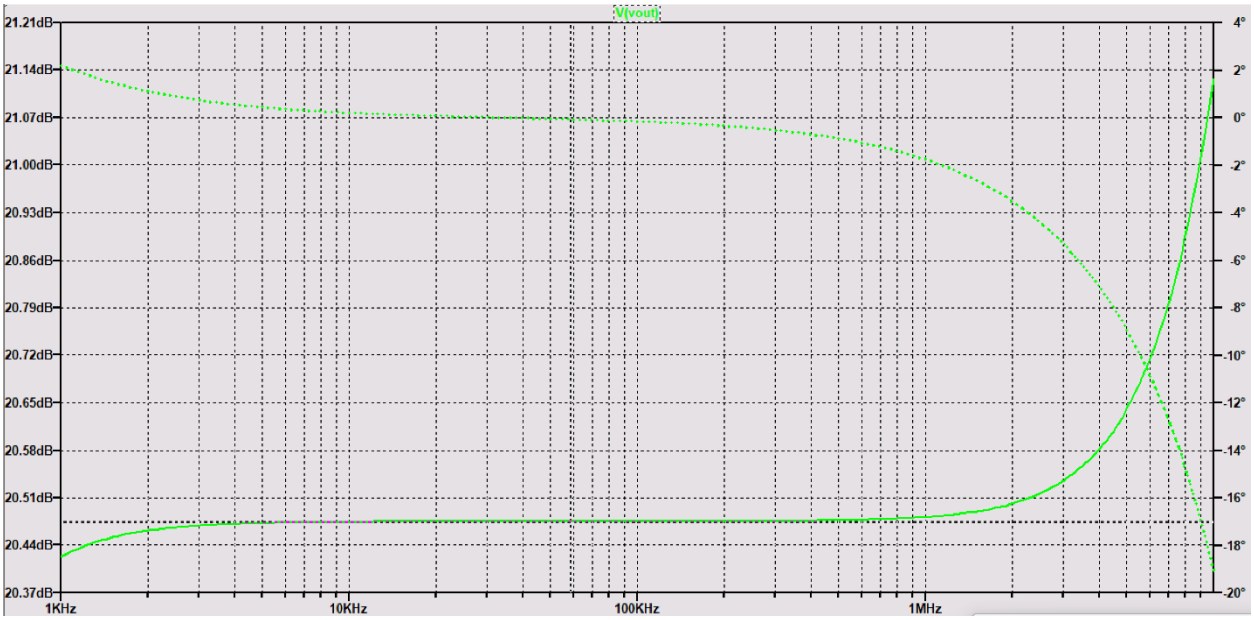


Figure 15: Gain plot when the load resistor is open-circuited.

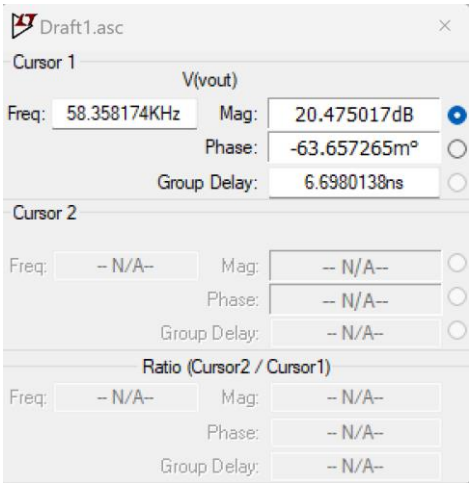


Figure 16: Gain value when the load resistor is open-circuited

Then, to reach a drop of 6dB, 4.7Ω is replaced for the load resistor. This is the closest standard resistor value to reach this.

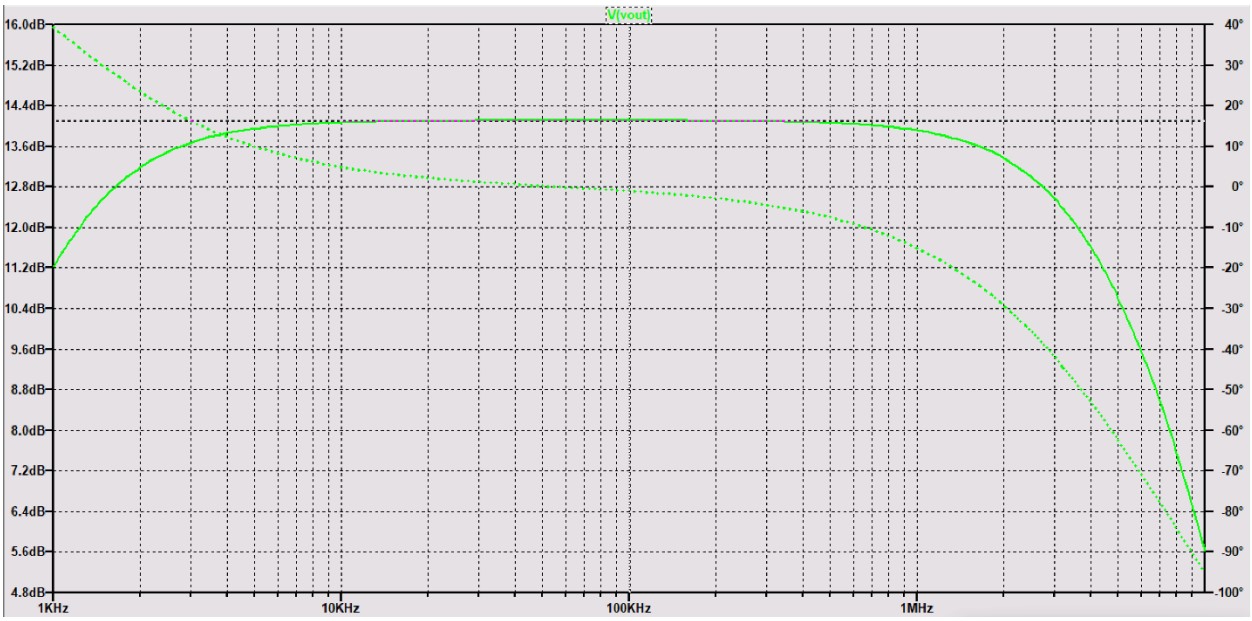


Figure 17: Gain plot when the load resistance is 4.7Ω.

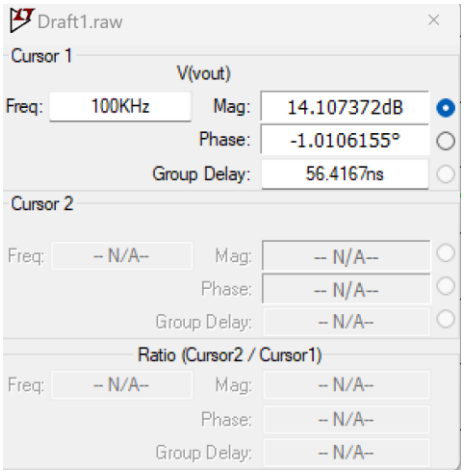


Figure 18: Gain value when the load resistance is 4.7Ω.

6. The phase margin of the open-loop system.

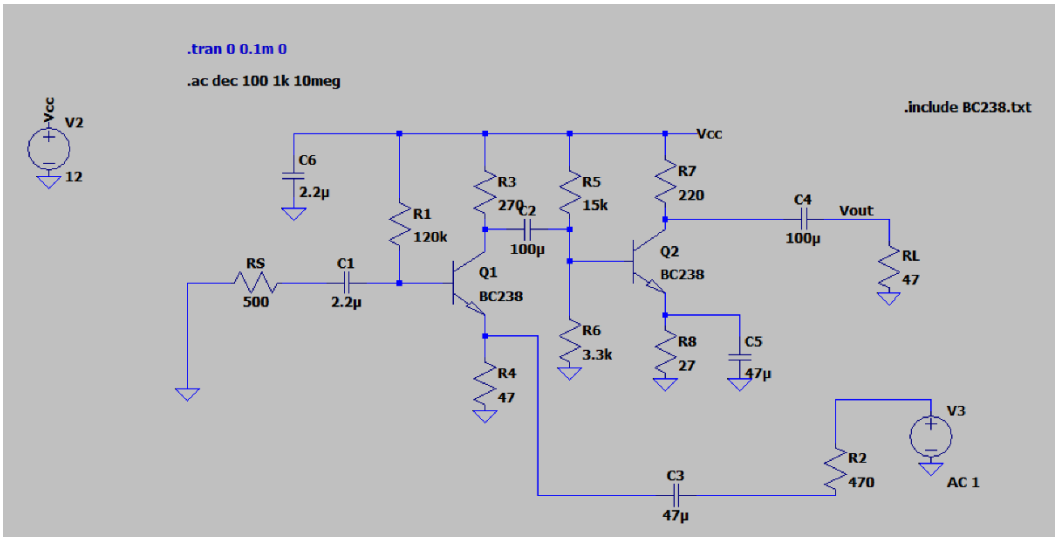


Figure 19: Open-loop circuit design on LTSpice.

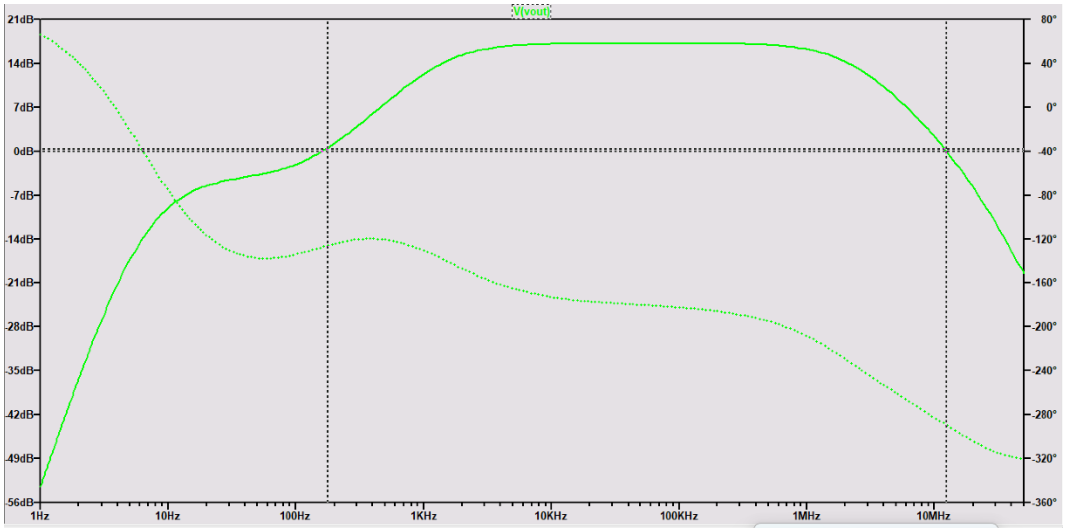


Figure 20: Gain plot of the open-loop circuit.

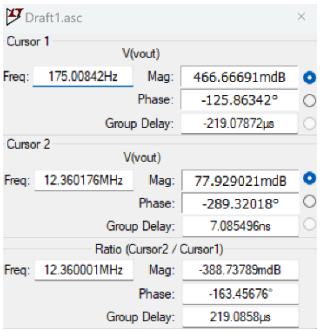


Figure 21: Phase difference of the gain is 163°

Hardware Part

In this part, the circuit will be implemented on hardware and the 6 specifications will be checked.

Here is the circuit implemented on a breadboard:

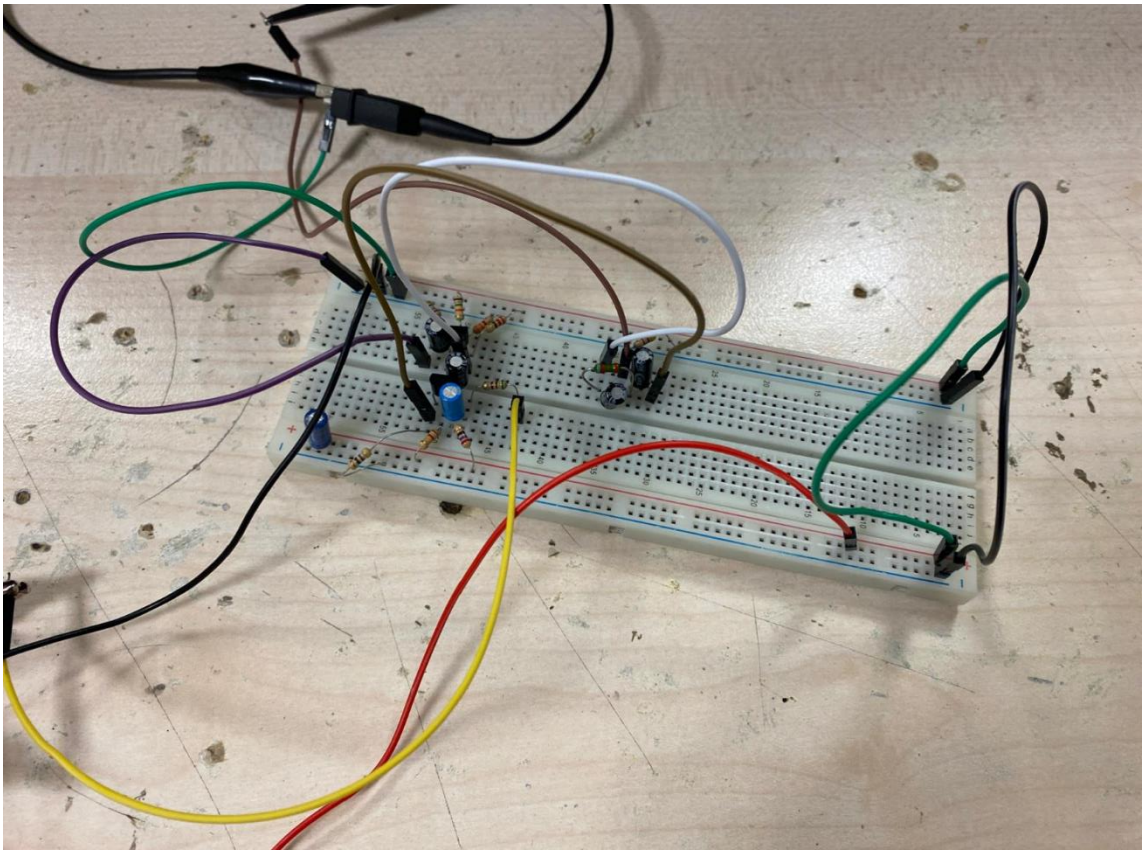


Figure 22: Circuit design implemented on a breadboard.

Specifications

1. **The current consumption is less than 70mA.**

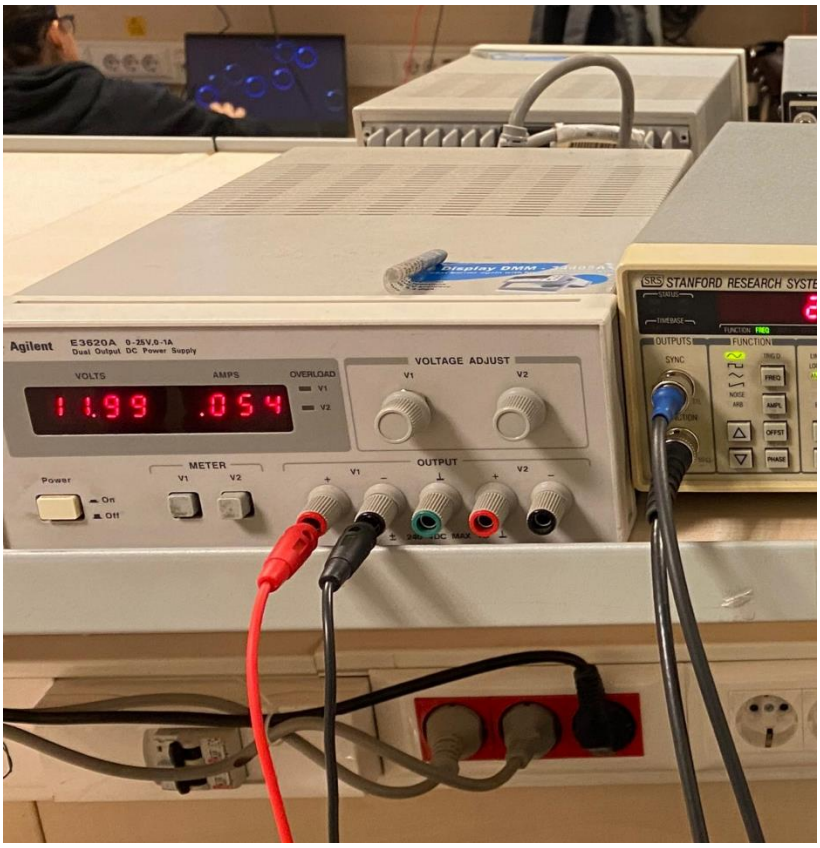


Figure 23: Current consumption 54mA.

We can clearly see that the current 54mA is less than 70mA so this criteria is satisfied.

2. **The small-signal bandwidth is at least 2KHz-2MHz while the mid-band gain is 20dB±0.5dB (by AC analysis)**

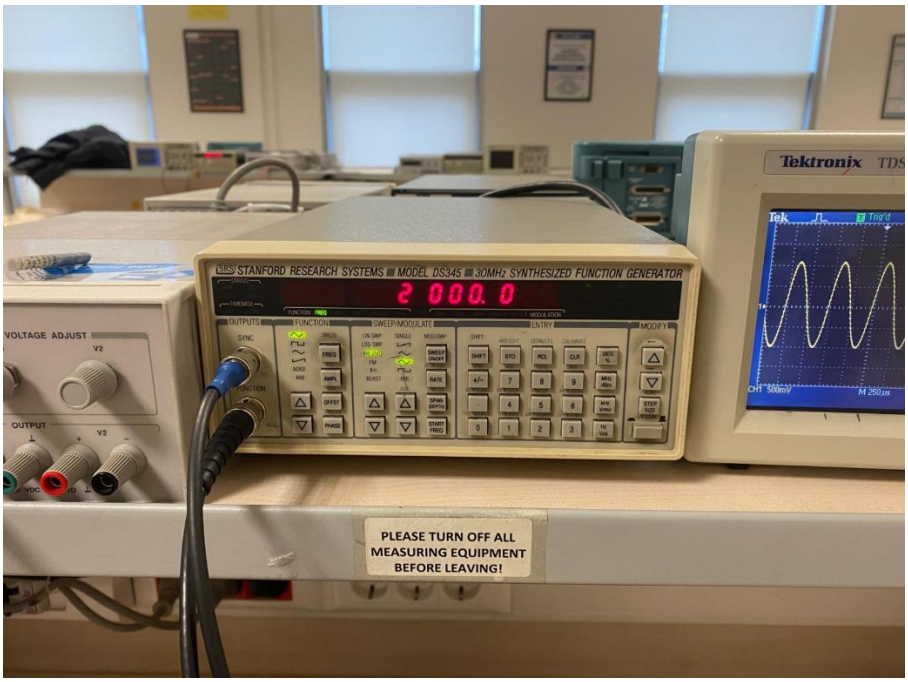


Figure 24: 2kHz Frequency.

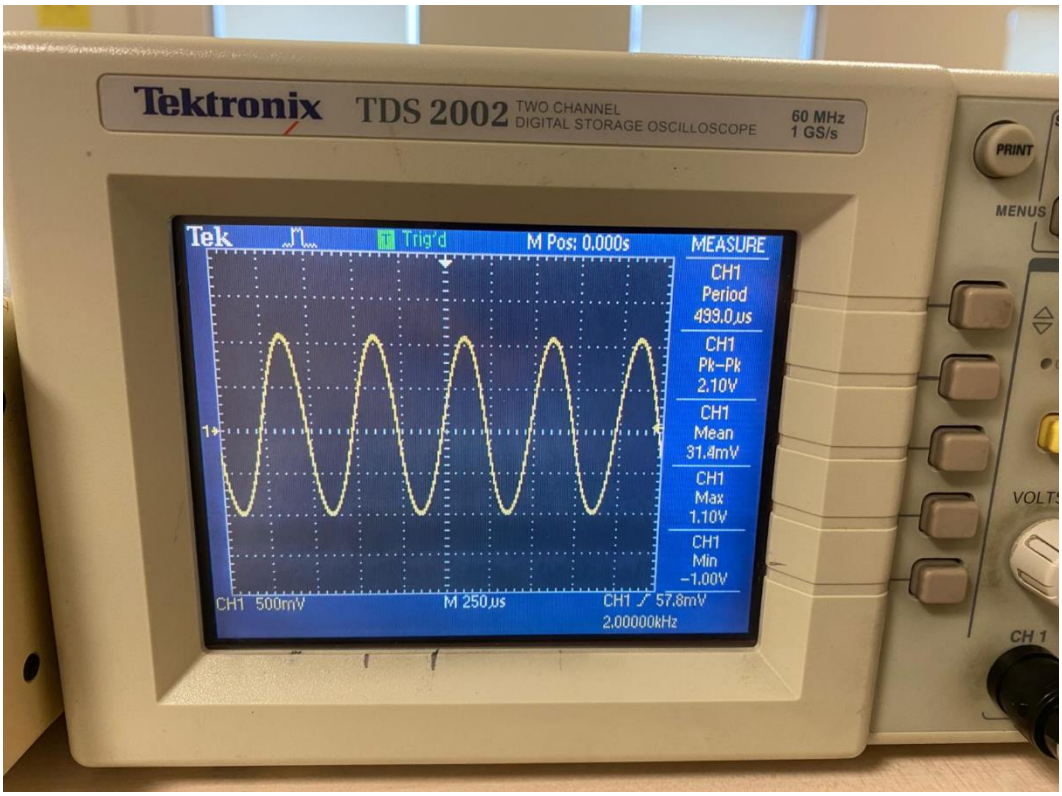


Figure 25: Output voltage at 2kHz frequency.

At 2kHz, we can calculate the gain by using this formula below:

$$20\log\left(\frac{2.10}{0.2}\right) = 20.42dB$$

Which is in our range.

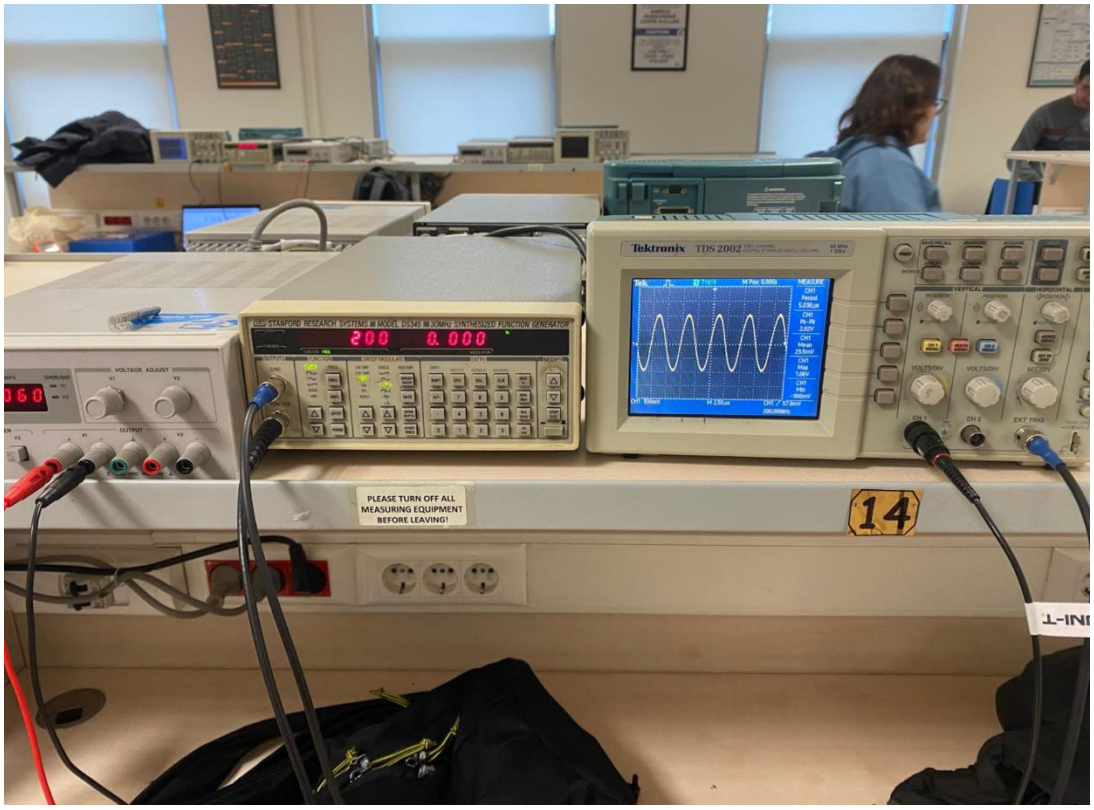


Figure 26: 200kHz Frequency

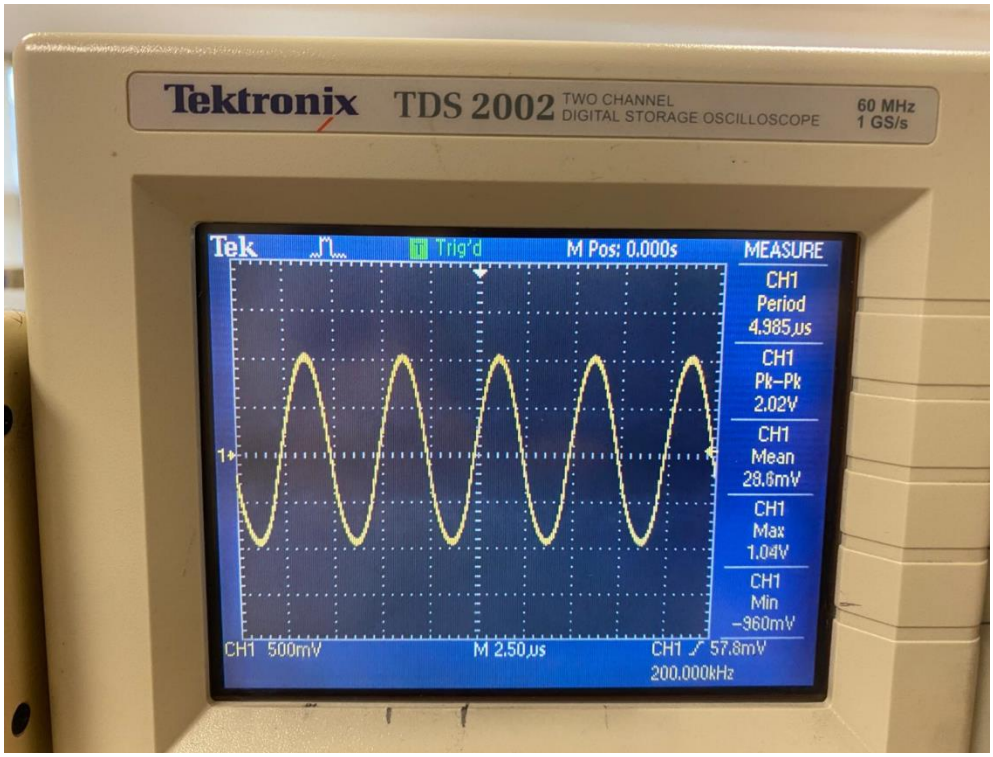


Figure 27: Output at 200kHz frequency.

At 200kHz, we can calculate the gain by using this formula below:

$$20\log\left(\frac{2.02}{0.2}\right) = 20.086dB$$

Which is almost 20dB. This is the wanted mid band gain value of this circuit.

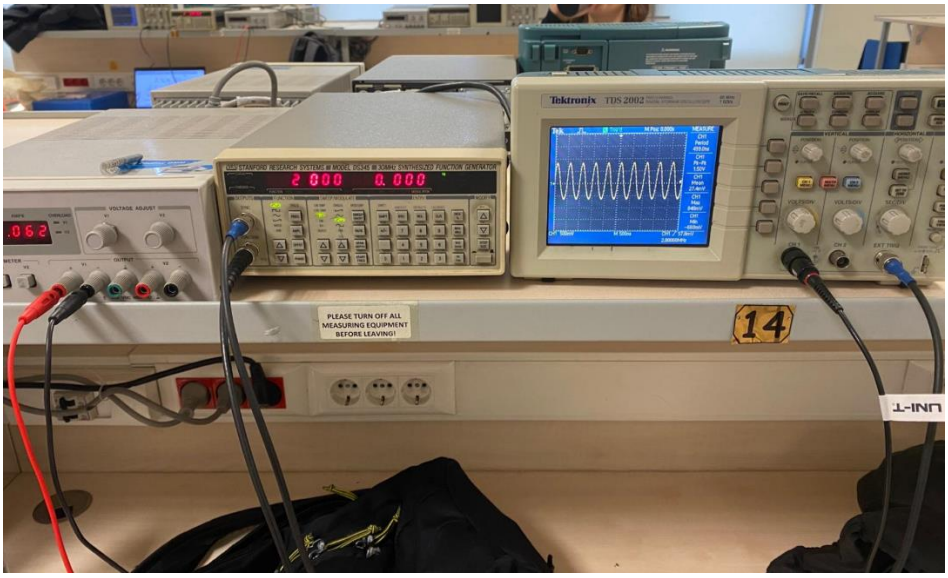


Figure 28: 2MHz frequency.

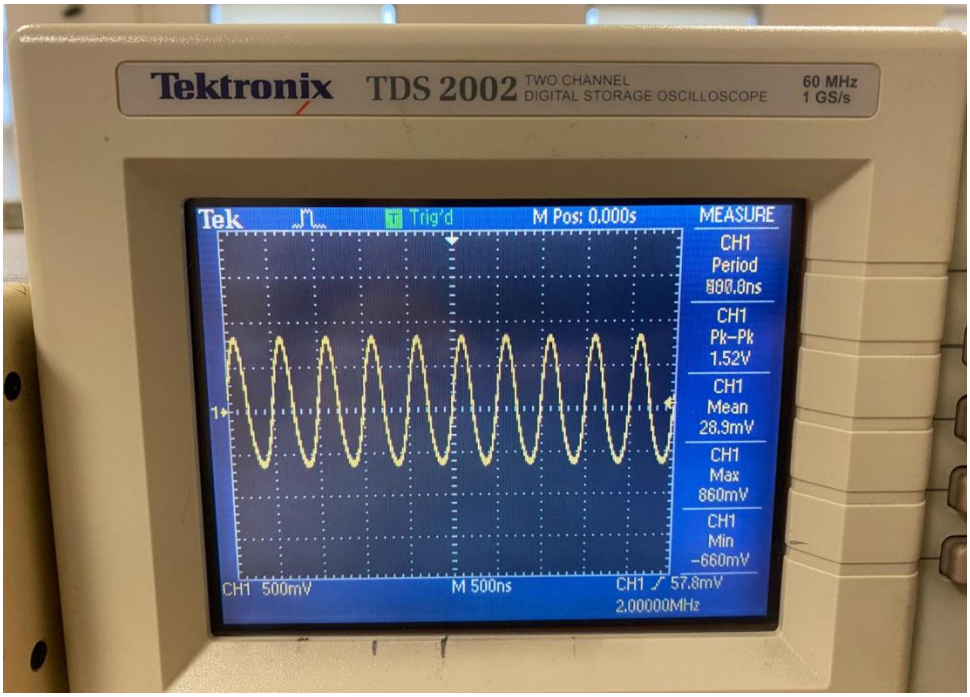


Figure 29: Output at 2MHz frequency.

Lastly for 2MHz, we can calculate the gain by using this formula below:

$$20\log\left(\frac{1.52}{0.2}\right) = 17.61dB$$

17.61dB is in the -3dB range so it is also satisfied.

3. The harmonic content of the output voltage is better than −30dBc with 0.1V peak input signal at 200KHz.

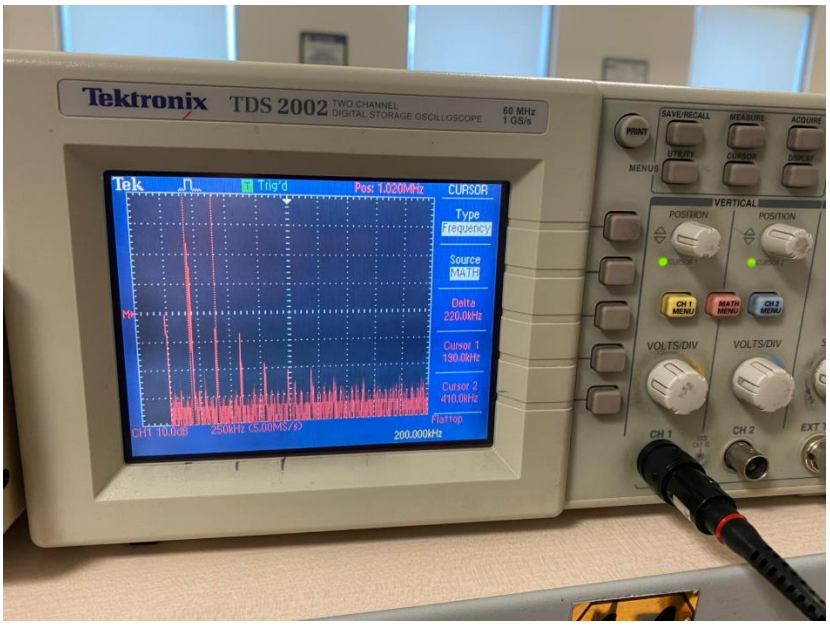


Figure 30: Harmonics at 200kHz and 400kHz.

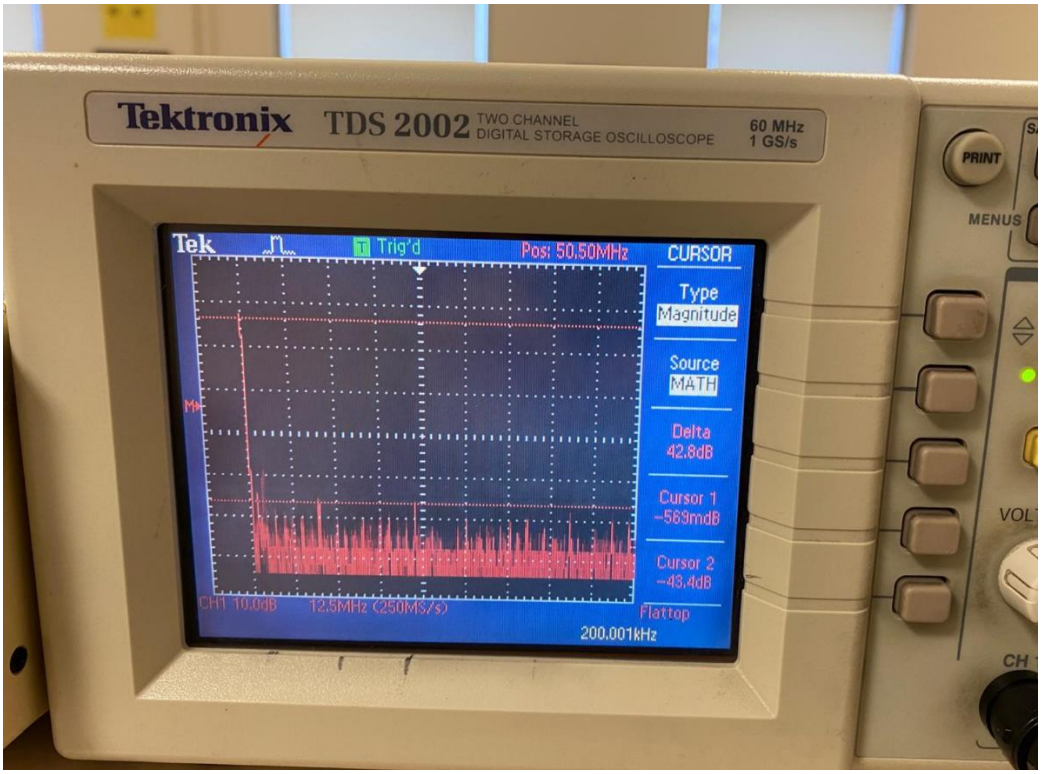


Figure 31: 42.8dB difference between the harmonics at 200kHz and 400kHz.

As it can be seen on the FFT plots, the difference between the first and the second harmonic is measured which is 42.8dB. That is a better than 30dB so this requirement is also satisfied.

4. **The small-signal input impedance of the amplifier at 200KHz (the adjusted value of RS in AC analysis until the voltage gain drops by 6dB compared to RS=0)**

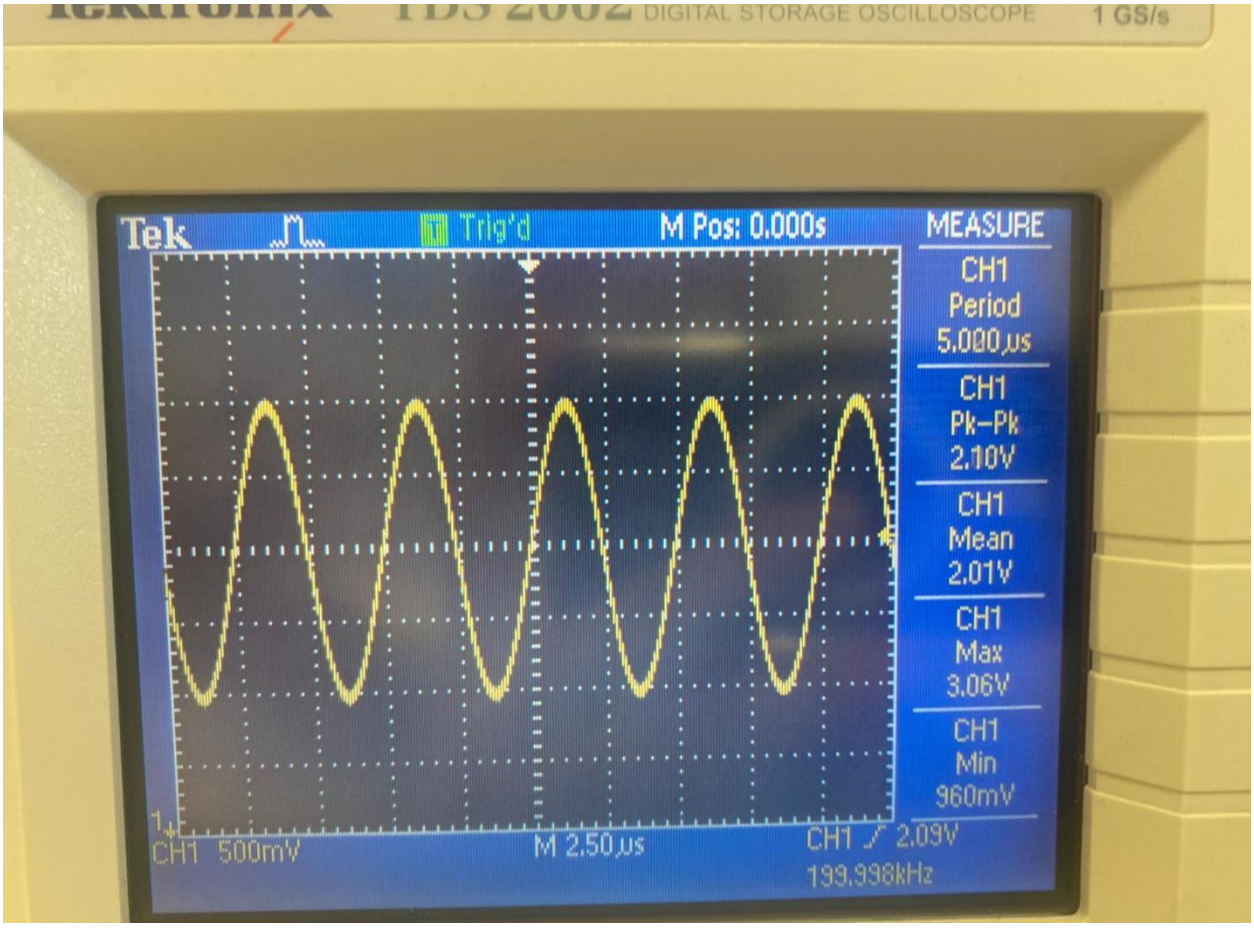


Figure 32: Vout = 2.1Vpp when Rs is shorted.

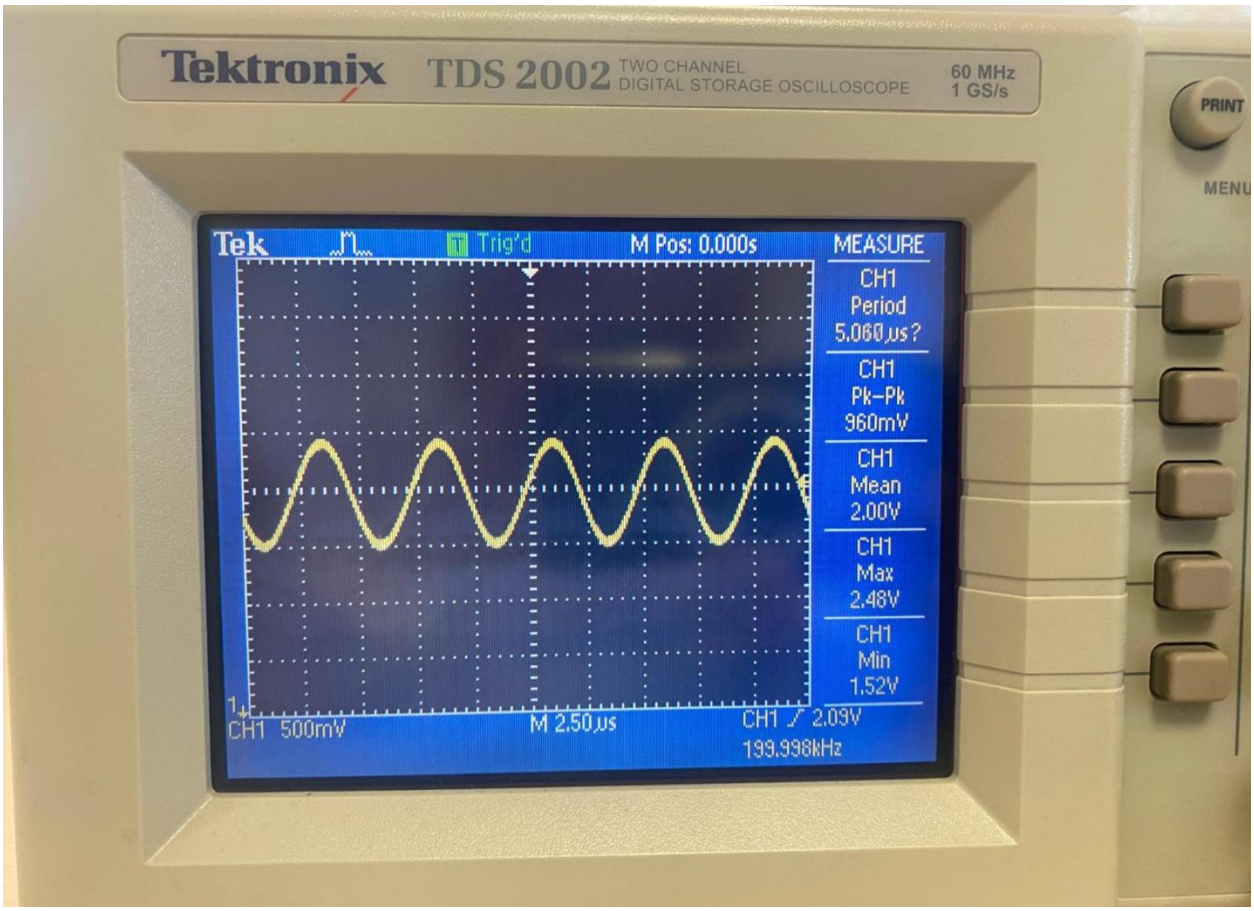


Figure 33: Vout=0.960Vpp when Rs=47k

When Rs is shorted, the calculated gain is 20.42dB by using the gain formula above. When Rs is changed to 47kΩ, our gain value changes to 13.62dB which is close to -6dB with Vout=960mVpp.

5. **The small-signal output impedance of the amplifier at 200KHz (the adjusted value of RL in AC analysis until the voltage gain drops by 6dB compared to RL=∞)**

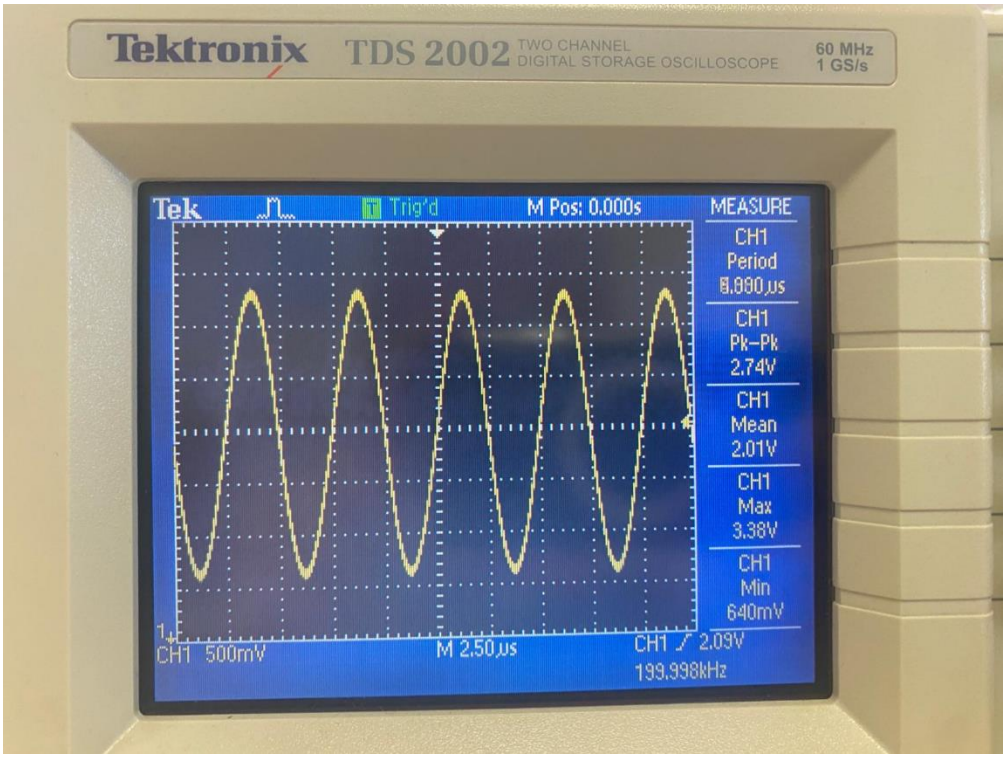


Figure 34: Vout = 2.74Vpp when the output is open circuited.

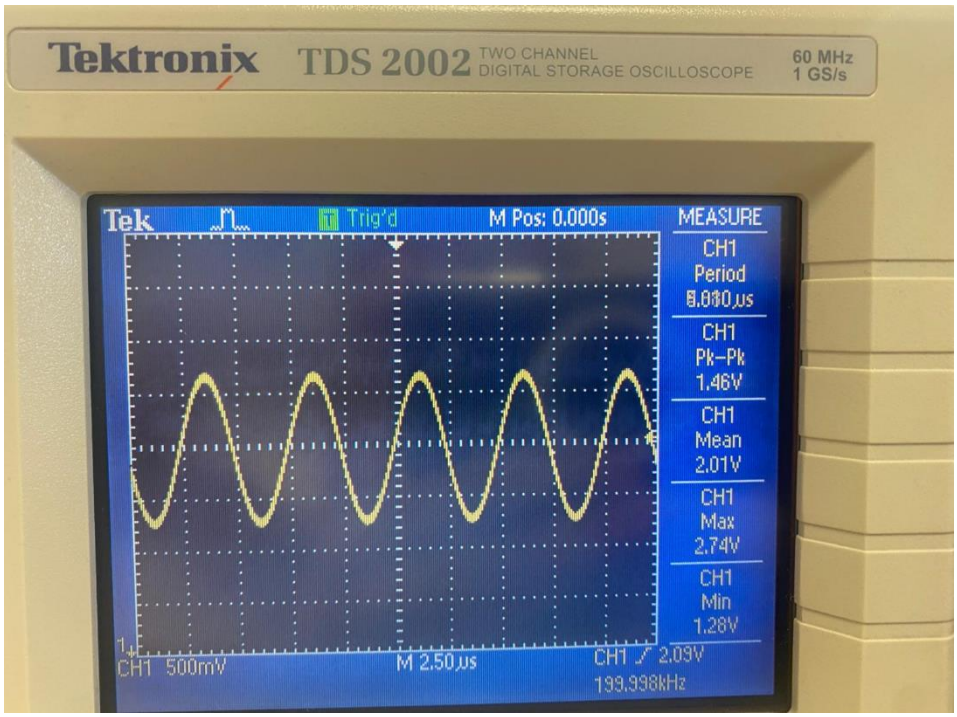


Figure 35: Vout=1.46Vpp when output resistor is changed to 4.7Ω

The gain when the output is shorted is 22.73dB with the gain formula above. When the output resistor is changed to 4.7Ω, the gain changes to 17.26dB. There is a decrease of almost 6dB which satisfies our last condition.

Conclusion

To conclude, all of the requirements were satisfied at the end with small errors. The general concept of wide-band amplifiers were learned and implemented on hardware. Breadboards were a big deal in this lab since they are not perfect tools to implement the circuit on. There were some misconnections but they were fixed at the end and a good result is observed. Type equation here.